

Moving fugacity asymptotics for distinct partitions

Uniform coefficient expansions and inverses for A291698 and A292304

1 October 2026

Abstract

For distinct partitions with size-dependent fugacity $u = n^\alpha$, this report proves a uniform all-orders Bessel expansion on every compact range $0 < a \leq \alpha \leq b$. An exact Jacobi transformation identifies finitely many conjugate exponential sectors, with rigorously separated local sector integrals and resonance-safe remainder bounds. The results imply the displayed leading equivalents for OEIS A291698 ($\alpha = 1$) and A292304 ($\alpha = 2$). A specified real continuation supports a controlled inverse, beginning with Lambert W and an explicit logarithmic correction. Exact computations expose a substantial preasymptotic oscillation: at $n = 20,000$, the principal approximation for $\alpha = 2$ still differs by about 5.6%. The scope is fixed sector count and fixed algebraic depth; no infinite Bessel-sector sum, growing-sector theorem, or numerical certification from unspecified asymptotic constants is claimed.

Reading guide. Sections 1–5 contain the coefficient and sector proofs. Section 6 gives the two OEIS specializations, Section 7 the inverse, and Sections 8–10 the computations, literature boundary, and further questions. The appendix records reproducibility details. The report makes no claim of publication priority or external peer review.

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1 Result and scope

Let

$$F(q, u) = \prod_{k \geq 1} (1 + uq^k), \quad a_n(\alpha) = [q^n]F(q, n^\alpha), \quad 0 < a \leq \alpha \leq b < \infty.$$

The endpoints a, b are fixed. Put

$$u = n^\alpha, \quad L = \log u, \quad A = -\text{Li}_2(-u), \quad c_1(u) = \frac{u}{12(1+u)},$$

$$N = n + c_1(u), \quad t = \sqrt{A/N}, \quad s = \sqrt{AN}.$$

Thus t is comparable to $\log n / \sqrt{n}$, s to $\sqrt{n} \log n$, uniformly in α . All square roots below are principal; for fixed integer j they have positive real part once n is large.

Let $D = u d/du$ and define

$$c_r(u) = \frac{B_{2r}}{(2r)!} D^{2r-1} \log(1+u), \quad \sum_{m \geq 0} h_m(u) z^m = \exp \left(\sum_{r \geq 2} c_r(u) z^{2r-1} \right)$$

as a **formal** power series. No convergence of the infinite Euler–Maclaurin series is asserted. Define

$$A_j = A - 2\pi^2 j^2 + 2\pi i j L, \quad T_{j,m} = \left(\frac{A_j}{N} \right)^{(m+1)/2} I_{m+1}(2\sqrt{N A_j}),$$

and

$$B_0 = (1+u)^{-1/2} t I_1(2s), \quad \Delta_j = 2s - 2\Re \sqrt{N A_j}.$$

1.1 Theorem 1 Principal sector expansion

For every fixed integer $M \geq 0$,

$$a_n(\alpha) = (1+u)^{-1/2} \sum_{m=0}^M h_m(u) t^{m+1} I_{m+1}(2s) + B_0 O_{a,b,M} \left(\frac{t^{M+1}}{u} + \exp\{-c/[t(1+L)^2]\} \right). \quad (1)$$

Here $c > 0$ depends only on a, b (indeed the underlying interval bound is absolute). The second error is smaller than every power of n , uniformly in α . Consequently it can be absorbed into the first error asymptotically. In particular, taking $M = 2$ gives

$$a_n(\alpha) = B_0 \{1 + O_{a,b}(t^3/u)\}. \quad (2)$$

The absorbed form of (2) can have a very late numerical onset. The explicit exponential remainder in (1) should be retained in practical work.

1.2 Theorem 2 Finite resonance sectors

For fixed nonnegative integers K, M , the stronger, finite-sector statement is

$$a_n(\alpha) = (1+u)^{-1/2} \sum_{|j| \leq K} (-1)^j \sum_{m=0}^M h_m(u) T_{j,m} + B_0 O_{a,b,K,M} \left(\frac{t^{M+1}}{u} + e^{-\Delta_{K+1}} \right). \quad (3)$$

The j and $-j$ terms are conjugates, so the displayed approximant is real. For each fixed $j \geq 1$,

$$\Delta_j \sim \frac{2\sqrt{2}\pi^4}{3} \frac{j^2 \sqrt{n}}{L^3} \sim \frac{2\pi^4 j^2}{3tL^2}. \quad (4)$$

Uniformity in (3) is for **fixed** K, M and α in a fixed positive compact interval. Neither K growing with n nor an infinite sum of Bessel sectors is asserted. Section 5 gives the finite-arc proof, including the choice of an auxiliary larger fixed sector cutoff. The exact local-sector version is stated separately in Section 5.4.

Formula (3) is an all-algebraic-order expansion inside finitely many exponentially small sectors. At any fixed M , its principal-sector algebraic remainder ultimately exceeds all retained exponential sectors; it is not a claim of exponentially precise hyperasymptotics with M growing with n . The exact local-sector formulation in Section 5.4 supplies a stronger, non-vacuous exponential separation before the algebraic expansions are truncated.

1.3 Provenance, status and reading conventions

[write] **Added 2 October 2026, batch 77.** *Provenance.* This report was built from one manuscript: batch 77, manuscript 34 of the repository's incoming-reports channel, archive `moving-fugacity-reproducibility.zip` (main file `moving-fugacity-report.tex`, 985 lines, dated 1 October 2026, with a 17-page PDF that is not shipped), arrival commit `096ee7b87`, placed in `4f11bc9c0`. The text from the title to the end of the scope checklist is the delivered manuscript, unchanged except for the label prefix `mfp`: and the notes marked [write]; no statement, proof, number or symbol of it was changed, and every section number the text cites by hand is still correct. The delivered proof note `theorem-and-proof.md` and the delivered `REPRODUCIBILITY.md` are shipped beside this article, byte-identical; the SHA-256 printed in Appendix A still matches the shipped `theorem-and-proof.md`. *Pin.* The delivery names no ProveIt revision and continues no repository report. *Status.* An unrefereed research manuscript (its author line is empty). Nothing in it is formalized in Lean or Rocq, and its placement in the repository confers no formal status. At intake and at this write all four delivered programs (Appendix A) were rerun on a copy, with outputs identical to the shipped records up to line endings; the first ten terms of both OEIS sequences were recomputed independently by a direct product expansion. These are numerical diagnostics; the proofs were not re-derived.

[write] **Reading conventions.** The manuscript reuses several letters; no symbol was renamed. Two readings need care. *The fugacity* $u = n^\alpha$ *is one weight per part, the same for every part of a partition of* n : a partition with m parts has weight $n^{\alpha m}$. It is *not* a part-dependent weight k^α on a part of size k , which is the different family $\prod_k (1 + k^\alpha q^k)$ (see the note at the end of Section 9). *The symbol* C_n *has two unrelated meanings*: in Section 2 it is the Bessel scale $C_n = A^{1/4} e^{2\sqrt{A}N} / (2\sqrt{\pi} N^{3/4} \sqrt{1+u})$, while in Sections 5.2–5.3 it is the bounded constant $C_n = A - L^2/2 \rightarrow \pi^2/6$.

symbol	meanings, by section
a, b	the endpoints $0 < a \leq \alpha \leq b$ of the compact α -range; $a_n(\alpha)$ is the sequence; a_j, b_j are $\Re\sqrt{A_j}, \Im\sqrt{A_j}$ in Section 5.3; $b_r(\nu)$ are Bessel coefficients (Section 2).
A	$A = -\text{Li}_2(-u)$; A_j its sector shifts; A_∞ its elementary part (Section 6); $\mathcal{A}_\alpha(x)$ the real continuation (13).
B	B_{2r} Bernoulli numbers; B_0 the principal scale; $\mathcal{B}_j$ the exact sector integrals (12c).
c	$c_r(u)$ endpoint coefficients ($c_1(u) = u/(12(1+u))$); c, c_0, c_J, c_j unnamed positive constants, and c_1 in (9) is such a constant, not $c_1(u)$.
v	$v = 1/u$, the small fugacity (Sections 5.1, 7); a summation index in $b_r(\nu)$ and in the residue series after (10).
y	$y = tx - L$ in Section 3; $y = \theta/t$ in Section 5.2; $y = \log X$ in Section 7.
z	the complex variable $q = e^{-z}$; but $z = y/(2\sqrt{2}\alpha)$ in (15).
m, r	m is the endpoint degree, and also the number of consecutive indices in (9); r indexes c_r , is the starting index in (9) and the Lagrange order in (17); r_0 in (15).
x	the real continuation variable of Section 7; an integration variable ($w = xz$) in Section 3.
D, E, g, H	$D = u d/du$ (and D_v); $E(x) = \log S_{K,M}(x) - g(x)$ in (17), while $E_n(\alpha)$ is the Sommerfeld expression of Section 6; $g(x) = \sqrt{2}\alpha\sqrt{x} \log x$, while g_d are generator coefficients (Section 2); $H = g^{-1}$, while H_{ex} is the amplitude (12b).
η	a small fixed constant in (5); $\eta(y) = E(H(y))$ in (17); the Dedekind- η transformation in Section 5.
R_K, R_P	the exact remainder of (12d); the Euler–Maclaurin remainder of (13a).
t, s, N, L	$t = \sqrt{A/N}$ (the saddle, not a time), $s = \sqrt{AN}$, $N = n + c_1(u)$, $L = \log u$.

2 Coefficient generator and ordinary relative expansion

The first endpoint coefficients are

$$c_2 = \frac{u(u-1)}{720(1+u)^3}, \quad c_3 = -\frac{u(u-1)(u^2-10u+1)}{30240(1+u)^5},$$

$$c_4 = \frac{u(u-1)(u^4-56u^3+246u^2-56u+1)}{1209600(1+u)^7}.$$

For each fixed $r \geq 2$, $c_r(u) = O_r(u^{-1})$. The h_m generator can be evaluated without symbolic exponentiation. Set $g_{2r-1} = c_r$ for $r \geq 2$ and all other $g_d = 0$; then

$$h_0 = 1, \quad mh_m = \sum_{d=1}^m dg_d h_{m-d}.$$

In particular,

$$h_1 = h_2 = h_4 = 0, \quad h_3 = c_2, \quad h_5 = c_3, \quad h_6 = c_2^2/2, \quad h_7 = c_4, \quad h_8 = c_2 c_3, \quad h_9 = c_5 + c_2^3/6.$$

For every fixed $m > 0$, $h_m = O_m(u^{-1})$. This last bound alone does not prove a remainder with $1/u$; that issue is addressed explicitly in Section 3.

The ordinary large-argument Bessel expansion is

$$I_\nu(2s) \sim \frac{e^{2s}}{\sqrt{4\pi s}} \sum_{r \geq 0} b_r(\nu) s^{-r}, \quad b_r(\nu) = \frac{(-1)^r}{r! 16^r} \prod_{v=1}^r [4\nu^2 - (2v-1)^2].$$

Thus the principal relative expansion is generated by the double series

$$\frac{a_n}{C_n} \sim \sum_{m,r \geq 0} h_m(u) t^m b_r(m+1) s^{-r}, \quad C_n = \frac{A^{1/4} e^{2\sqrt{AN}}}{2\sqrt{\pi} N^{3/4} \sqrt{1+u}}.$$

Its first Bessel terms are

$$1 - \frac{3}{16s} - \frac{15}{512s^2} - \frac{105}{8192s^3}.$$

The first new endpoint contribution is $c_2 t^3 I_4(2s)/I_1(2s)$, of order $n^{-\alpha-3/2} \log^3 n$. Keeping exact Bessels is preferable to expanding them numerically. For any desired fixed algebraic precision n^{-R} , choose M and the Bessel expansion depth sufficiently large; the log factors are accounted for by choosing strict inequalities in the depth.

3 Euler–Maclaurin and its uniform remainder

Write $f_u(w) = \log(1 + ue^{-w})$, with the branch continued from the positive real axis. Fix a small $\eta > 0$. In the strip-like sector

$$\Re z = t > 0, \quad |\Im z| \leq \eta t / (1 + L), \quad (5)$$

the path $w = xz$, $x \geq 0$, stays uniformly away from the zeros of $1 + ue^{-w}$. Near $tx = L$ its argument is bounded by η ; away from that transition its modulus is either much larger or much smaller than one. For each integer $r \geq 2$,

$$\int_0^\infty |f_u^{(r)}(xz)| dx \leq C_r/t.$$

One direct verification uses $y = tx - L$: the derivatives are rational functions of ue^{-xz} , bounded by $C_r e^{-|y|}$ outside a fixed y interval and bounded uniformly inside it. The integral term is A/z , by the analytic antiderivative or contour rotation. Euler–Maclaurin through index P therefore gives

$$\log F(e^{-z}, u) = A/z - \frac{1}{2} \log(1+u) + \sum_{r=1}^P c_r(u) z^{2r-1} + O_P(t^{2P-1}), \quad (6)$$

uniformly in (5). The generic remainder here does **not** contain $1/u$.

To infer (1), fix M and take P with $2P > M + 2 + 2b$. Then

$$t^{2P-1} = O_{a,b,M,P}(t^{M+1}/u),$$

because $u \leq n^b$ and t is comparable to $\log n / \sqrt{n}$. Each discarded endpoint term with degree at least $M+1$ is individually $O(t^{M+1}/u)$, since $c_r = O(u^{-1})$. Exponentiating and collecting coefficients through degree M gives

$$F(e^{-z}, u) = \frac{e^{A/z + c_1 z}}{\sqrt{1+u}} \left\{ \sum_{m=0}^M h_m(u) z^m + O(t^{M+1}/u) \right\}. \quad (7)$$

For $M = 0, 1, 2$ the first nonzero omitted endpoint degree is 3, yielding the sharper $O(t^3/u)$ if desired. An alternative derivation using the exact identity in Section 5 obtains the $1/u$ factor directly from the small-fugacity factor's derivative norm.

4 Minor arcs and Bessel transfer

The Cauchy integral on $q = e^{-t+i\theta}$ is

$$a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{n(t-i\theta)} F(e^{-t+i\theta}, u) d\theta.$$

Set $p_k = ue^{-kt}/(1 + ue^{-kt})$. Exactly,

$$\left| \frac{F(e^{-t+i\theta}, u)}{F(e^{-t}, u)} \right| \leq \exp \left\{ -2 \sum_{k \geq 1} p_k (1 - p_k) \sin^2(k\theta/2) \right\}. \quad (8)$$

On the consecutive integer interval $|kt - L| \leq 1$ there are m comparable to $1/t$ indices, all with $p_k(1 - p_k) \geq c_0 > 0$. The following elementary bound, for $m \geq 2$, is independent of the starting index:

$$\sum_{k=r}^{r+m-1} \sin^2(k\theta/2) \geq c_1 m \min\{m^2\theta^2, 1\}, \quad |\theta| \leq \pi. \quad (9)$$

Indeed the minimum over the initial phase is at least

$$m/2 - \frac{1}{2} \left| \frac{\sin(m\theta/2)}{\sin(\theta/2)} \right|.$$

For $m|\theta| \leq 1$, Taylor bounds on $\sin$ give a fixed multiple of $m^3\theta^2$. For $1 \leq m|\theta| \leq C$, compactness of $|\sin x|/|x|$ away from zero gives a fixed fraction of m , uniformly as m grows. For $m|\theta| \geq C$, choose $C > 2\pi$ and use $|\sin(\theta/2)| \geq |\theta|/\pi$. This proves (9), with finitely many small $m \geq 2$ absorbed by reducing the constant. The exclusion $m = 1$ is necessary, and irrelevant here because m is comparable to $1/t$ and tends to infinity.

For $|\theta| \geq \eta t/(1+L)$, (8)-(9) give a loss at least $c/[t(1+L)^2]$. Polynomial factors from normalization can be absorbed by reducing c . Since

$$1/[t(1+L)^2] \asymp \sqrt{n}/(\log n)^3,$$

these arcs are uniformly beyond all algebraic orders. This argument explicitly controls the near-resonances $\theta \approx 2\pi/\sqrt{2n}$; a fixed-fugacity variance bound alone is not a substitute.

On the remaining arc, insert (7). For $z = t + i\theta$,

$$\Re(A/z + Nz) = 2s - \frac{s(\theta/t)^2}{1 + (\theta/t)^2}.$$

Consequently integrating a uniform $O(t^{M+1}/u)$ error costs only $O(e^{2s}t/\sqrt{s})$ times that error, the scale of the main Bessel integral.

For each fixed m , deform the truncated vertical segment to the nearby arc of $|z| = t$, adding short radial connectors. At an endpoint $\theta = \eta t/(1+L)$, all connectors lose at least $cs/(1+L)^2$, comparable to c/t , in real exponent. Completing the circle loses the same amount. The exact residue is

$$\frac{1}{2\pi i} \oint_{|z|=t} z^m e^{A/z + Nz} dz = (A/N)^{(m+1)/2} I_{m+1}(2\sqrt{AN}). \quad (10)$$

For completeness, expand both exponentials: the residue is $\sum_{v \geq 0} A^{v+m+1} N^v / ((v+m+1)!v!)$, which is exactly the Bessel series. Equations (7)-(10) prove Theorem 1.

5 Exact Jacobi decomposition and the finite-sector proof

Let $q = e^{-z}$, $\Re z > 0$, $u > 1$, $L = \log u$, $Q = e^{-4\pi^2/z}$. With $(a; q)_\infty = \prod_{r \geq 0} (1 - aq^r)$, the following is an **exact identity**:

$$F(e^{-z}, u) = \frac{u^{-1/2} \exp\{(L^2/2 + \pi^2/6)/z + z/12\}}{(Q; Q)_\infty (-1/u; e^{-z})_\infty} \sum_{j \in \mathbb{Z}} (-1)^j \exp\{(-2\pi^2 j^2 + 2\pi i j L)/z\}. \quad (11)$$

Proof: Jacobi's triple product gives

$$F(q, u) (-1/u; q)_\infty (q; q)_\infty = \sum_{m \in \mathbb{Z}} u^m q^{m(m+1)/2}.$$

Complete the square and apply Gaussian Poisson summation. The bilateral sum becomes

$$\sqrt{2\pi/z} e^{L^2/(2z) - L/2 + z/8} \sum_{j \in \mathbb{Z}} (-1)^j e^{(-2\pi^2 j^2 + 2\pi i j L)/z}.$$

The η transformation is

$$(e^{-z}; e^{-z})_\infty = \sqrt{2\pi/z} e^{-\pi^2/(6z) + z/24} (Q; Q)_\infty.$$

Division proves (11), first for $z > 0$ and then throughout $\Re z > 0$ by analytic continuation, with consistent square-root branches.

[write] **Added 5 October 2026, batch 100: a q -dependent fugacity.** Identity (11) holds for every fixed $u > 1$; at real $z = t$ it may be applied with a fugacity that depends on q . The report **a126348-stable-hilbert-series** (bundle Report 84, placed in **36571ae0e**) treats $\prod_{k \geq 1} (1 + q^k/(1-q))$, OEIS A126348, the case $u = 1/(1 - e^{-t})$: with $\log u = \lambda$ there, (11) becomes its exact factorization $F(e^{-t}) = e^{K(t)} \Theta / (Q; Q)_\infty$ (its Theorem 2.1, label **shs:thm:modular**) term by term, by the same proof. That manuscript did not cite this report (its pin predates the placement **4f11bc9c0**); the credit is recorded there. Its coefficient regime is different: the fugacity $1/(1-q)$ is large only near $q = 1$, so its minor-arc bound leaves

no resonance sectors, in contrast with the finitely many conjugate sectors of Theorem 2 here; and $\log u$ is about $\frac{1}{2} \log n$ at its saddle, so it does not bear on question 10.3 (fugacity with $\log u$ comparable to $n^{1/6}$).

5.1 Uniform local amplitude over any fixed number of sectors

Fix $C > 0$ and allow $|\Im z| \leq Ct/L$, $\Re z = t$. Unlike (6), C need not be small. Apply Euler–Maclaurin to $\log(-1/u; e^{-z})_\infty$, including its $k = 0$ endpoint. Its fugacity $v = 1/u$ tends to zero. For $r \geq 1$, the corresponding derivative L^1 norm is $O_r(1/(ut))$, because $|ve^{-xz}| \leq v$ and the denominator is at least $1 - v$. Hence its remainder is directly $O(t^{2P-1}/u)$, throughout this larger sector.

The identity

$$A = \frac{1}{2}L^2 + \pi^2/6 + \text{Li}_2(-1/u)$$

and the relations $1/12 - c_1(1/u) = c_1(u)$, $c_r(1/u) = -c_r(u)$ for $r \geq 2$ transform (11) into

$$F(e^{-z}, u) = \frac{1}{\sqrt{1+u}} \left\{ \sum_{m=0}^M h_m(u) z^m + O(t^{M+1}/u) \right\} \sum_{j \in \mathbb{Z}} (-1)^j e^{A_j/z + c_1 z} \{1 + O(e^{-c/t})\}. \quad (12)$$

The last factor comes from $(Q; Q)_\infty$, since $\Re(1/z)$ is comparable to $1/t$ for fixed C and L tending to infinity. For a fixed desired M choose $2P - 1 \geq M + 1$. The estimate on the brace is uniform in z ; its error multiplies the theta sum, so estimates below use the **sum of the absolute values** of the theta terms, never division by a possibly vanishing theta sum.

5.2 Isolating and completing finitely many saddle arcs

Fix an auxiliary integer $J > K$, to be chosen sufficiently large below. Restrict initially to

$$|\theta| \leq (2J + 1)\pi t/L.$$

Split this interval at $\theta = (2j + 1)\pi t/L$, so the j -th cell is centered at $2\pi j t/L$ (reverse the sign if using $z = t - i\theta$). More directly, put $y = \theta/t$ and $C_n = A - L^2/2$. The exact completed-square identity is

$$\Re\{A_j/(t + i\theta) + N(t + i\theta)\} = 2s - \frac{C_n y^2}{t(1 + y^2)} - \frac{2\pi^2}{t(1 + y^2)} \left(j - \frac{Ly}{2\pi}\right)^2. \quad (12a)$$

On the restricted interval, $|Ly/(2\pi)| \leq J + 1/2$. Thus every $|j| \geq J + 1$ has distance at least $1/2$ in the final square, and summing this Gaussian tail gives $O(e^{2s - c_J/t})$. This justifies replacing the infinite theta sum by the finite sum $|j| \leq J$ on this interval. One may integrate each of these finitely many terms on the entire restricted interval; assigning individual cells is unnecessary. The identity also shows that the sum of absolute values of the finite terms never has an exponentially large maximum relative to e^{2s} .

For each $|j| \leq J$ the saddle is $z_j = \sqrt{A_j/N}$. It satisfies

$$\Re z_j = t\{1 + O_J(L^{-4})\}, \quad \Im z_j = 2\pi j t/L + O_J(t/L^3).$$

Move the local vertical segment through $\Re z_j$, then to the circle $|z| = |z_j|$. All its endpoints are separated from $\arg z_j$ by a fixed multiple of $1/L$, so the joining contours and omitted circular arcs lose c_J/t in real exponent. On that circle, writing $\beta = \arg z_j$, the phase is

$$A_j/z + Nz = 2\sqrt{NA_j} \cos(\phi - \beta),$$

and its real part is maximal at $\phi = \beta$. Thus the completed integral is precisely $T_{j,m}$, with exponentially negligible endpoint error. The Gaussian absolute integral is $O(te^{2\Re\sqrt{NA_j}/\sqrt{s}})$, uniformly for $|j| \leq J$. Summing the amplitude remainders gives $B_0 O(t^{M+1}/u)$. One can deform the analytic remainder along with the exact small-fugacity amplitude before truncating. Alternatively, on the original vertical line the maximum real exponent for sector j is $s + (\Re\sqrt{A_j})^2/t$, which is at most $2s$; its local absolute-integral width is comparable to $t/\sqrt{s}$, uniformly for the finitely many j . No division by the theta sum or exponentially growing relative-error factor is used.

A fully mechanical version of the contour argument can integrate each of the finitely many exponentials on the entire restricted vertical interval, then join to its saddle circle. Their endpoint losses are at least c_J/t because every retained saddle stays a distance comparable to t/L from the two outer endpoints. Terms whose centers lie just outside the interval have only endpoint Gaussian tails. This avoids making any assertion about dominance exactly at the cell boundaries.

5.3 Bounding the rest without an infinite Bessel sum

Let $C_n = A - L^2/2$. Then C_n tends uniformly to $\pi^2/6$ and is bounded between two positive constants. Since

$$A_j = \frac{1}{2}(L + 2\pi i j)^2 + C_n,$$

the real part of $\sqrt{A_j}$ is positive and decreases with $|j|$. One way to see this is to write $a_j = \Re\sqrt{A_j}$, $b_j = \Im\sqrt{A_j}$; the equations $a_j b_j = \pi j L$ and $a_j^2 - b_j^2 = A - 2\pi^2 j^2$ show that a_j decreases from $\sqrt{A}$ to $L/\sqrt{2}$ when $C_n > 0$. Equivalently differentiate the explicit formula $a_j^2 = (|A_j| + A - 2\pi^2 j^2)/2$. It follows that Δ_j increases with $|j|$. Expanding the square root for fixed j gives (4).

Each of the finitely many omitted sectors $K < |j| \leq J$ is therefore $O(B_0 e^{-\Delta_{K+1}})$. Outside the enlarged interval, (8)-(9) give

$$|\text{outer contribution}|/B_0 \leq n^{C_0} \exp\{-c_0(2J+1)^2/[tL^2]\}$$

for all sufficiently large n ; the eventual condition $(2J+1)\pi/L < 1$ ensures the small-angle branch of (9) applies at the boundary, and its monotone lower bound handles all larger angles. Choose J , fixed in n , so $c_0(2J+1)^2$ exceeds twice an eventual upper bound for $\Delta_{K+1} t L^2$. That upper bound exists by (4). The polynomial factor is then absorbed, giving $O(B_0 e^{-\Delta_{K+1}})$. All connector and modular errors $e^{-c_J/t}$ are much smaller than this rate. This proves (3), without any infinite inversion or growing- K assertion.

5.4 Exact sector functions and a genuinely sectorwise statement

To distinguish an exponential sector before truncating a much larger principal algebraic series, define the analytic amplitude

$$H_{\text{ex}}(z, u) = \frac{\sqrt{1 + 1/u} \exp\{-\text{Li}_2(-1/u)/z + c_1(1/u)z\}}{(-1/u; e^{-z})_{\infty}}. \quad (12b)$$

It is analytic and nonzero for $\Re z > 0$, $u > 1$. Its endpoint expansion is exactly $\sum_m h_m z^m$. For each fixed integer j and sufficiently large n , let Γ_j be the vertical segment $z = t + i\theta$ with

$$(2j - 1)\pi t/L \leq \theta \leq (2j + 1)\pi t/L,$$

oriented upward, and set

$$\mathcal{B}_j(n, u) = \frac{(-1)^j}{2\pi i \sqrt{1+u}} \int_{\Gamma_j} H_{\text{ex}}(z, u) e^{A_j/z + Nz} dz. \quad (12c)$$

These are explicit exact local integrals, not infinite formal series. Define R_K by the exact identity $R_K = a_n - \sum_{|j| \leq K} \mathcal{B}_j$. Then, for fixed K ,

$$a_n(\alpha) = \sum_{|j| \leq K} \mathcal{B}_j(n, n^\alpha) + B_0 O_{a,b,K}(e^{-\Delta_{K+1}}). \quad (12d)$$

For each fixed j and M ,

$$\mathcal{B}_j(n, u) = \frac{(-1)^j}{\sqrt{1+u}} \sum_{m=0}^M h_m(u) T_{j,m} + B_0 e^{-\Delta_j} O_{a,b,j,M}(t^{M+1}/u). \quad (12e)$$

For $j = 0$ use $\Delta_0 = 0$. Proof of (12d): use the auxiliary J from Section 5.3. Inside its full arc, (11) is exactly the sum of the H_{ex} sector integrands times $(Q; Q)_{\infty}^{-1}$. The modular factor differs from 1 by $e^{-c/t}$. For the j -th exponential, the contribution outside its own Γ_j but inside the full arc is $e^{-c_j/t}$ times the principal scale, by (12a), because $|j - Ly/(2\pi)| \geq 1/2$. All infinitely many exterior θ indices have the same summable Gaussian bound. This leaves precisely the finite sum of exact local integrals $|j| \leq J$ plus $e^{-c_j/t}$ errors. Complete each fixed local integral to its saddle contour to bound its modulus by $O(B_0 e^{-\Delta_j})$; then drop $K < |j| \leq J$ and use the already established outer-arc estimate. This proves (12d).

For (12e), first deform the exact H_{ex} integrand to the short saddle-circle arc through z_j , keeping the entire deformation in $\Re z$ comparable to t and $|\Im z| \leq C_j t/L$. Apply and truncate the uniform small-fugacity Euler–Maclaurin expansion there. Only the resulting polynomial integrands are then completed around the full circle and evaluated by (10); H_{ex} is never continued into $\Re z \leq 0$. On the short saddle arc the Gaussian absolute-integral scale is $B_0 e^{-\Delta_j}$, so the analytic amplitude remainder has the required sectorwise bound. Endpoint and connector errors lose c_j/t in real phase and are smaller than t^{M+1}/u relative to the sector scale. This proves (12e). The order of these steps matters: on the original vertical line its maximum real phase exceeds $2\Re \sqrt{N A_j}$ by $(\sqrt{A} - \Re \sqrt{A_j})^2/t$, which need not be bounded. Estimating the sectorwise error there would be insufficient.

Thus the exponential error in (12d) is genuine before truncating the principal sector, and (12e) gives the algebraic expansion separately within every fixed sector. Numerically evaluating the exact local integrals is an alternative to using finite- M Bessel models; no implementation of certified integral quadrature is claimed in the accompanying code.

6 OEIS specializations and the fugacity threshold

The dilogarithm identity gives

$$A = A_\infty + \text{Li}_2(-1/u), \quad A_\infty = \frac{1}{2}\alpha^2 \log^2 n + \pi^2/6,$$

and $\text{Li}_2(-1/u) = -1/u + O(u^{-2})$. Therefore the fully elementary Sommerfeld expression

$$E_n(\alpha) = \frac{A_\infty^{1/4}}{2\sqrt{\pi} n^{\alpha/2+3/4}} e^{2\sqrt{nA_\infty}}$$

satisfies $a_n/E_n \rightarrow 1$ **if and only if** $\alpha \geq 1/2$, for fixed $\alpha > 0$. At $\alpha = 1/2$ the omitted exponent still tends to zero, but only at rate $1/\log n$. For $0 < \alpha < 1/2$,

$$\log \frac{a_n}{E_n} \sim -\frac{\sqrt{n}}{u\sqrt{A_\infty}} \sim -\frac{\sqrt{2} n^{1/2-\alpha}}{\alpha \log n},$$

so a_n/E_n tends to zero. This is a relative-asymptotic issue, not a small change to the leading logarithmic scale.

For $\alpha = 1$, the theorem proves the conjectural equivalent displayed for A291698:

$$a_n \sim \frac{(\log^2 n + \pi^2/3)^{1/4}}{\sqrt{\pi}(2n)^{5/4}} \exp \sqrt{2n(\log^2 n + \pi^2/3)}.$$

For $\alpha = 2$ it proves the displayed A292304 equivalent:

$$a_n \sim \frac{(2\log^2 n + \pi^2/6)^{1/4}}{2\sqrt{\pi}n^{7/4}} \exp\{2\sqrt{n(2\log^2 n + \pi^2/6)}\}.$$

Keeping A exact, N shifted, and the factor $\sqrt{1+u}$ exact is the clean route to the uniform theorem and its inverse.

7 Real continuation and inversion

A sequence has no unique real analytic continuation. To make the inverse unambiguous, specify the following **saddle-Fourier continuation**, for real x sufficiently large:

$$\mathcal{A}_\alpha(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{x(t(x)-i\theta)} F(e^{-t(x)+i\theta}, x^\alpha) d\theta, \quad (13)$$

where A, N, t are defined as above with n replaced by x . This integral is real, and equals $a_n(\alpha)$ at every positive integer $x = n$. This choice is a convention; no claim of uniqueness among continuations is made.

The preceding major/minor estimates work for real x without alteration, since $|e^{-ix\theta}| = 1$. In particular (13) is positive eventually. They can also be differentiated a fixed number of times: differentiating products introduces fixed sums of k powers and factors u'/u , t' , and θ , all bounded

by polynomial factors in x and $\log(x)$; removing a fixed number of product factors changes the minor-arc damping by at most a fixed constant. In the main arc differentiate the small-fugacity Euler–Maclaurin representation, taking additional endpoint depth as needed. More explicitly, let $R_P(x, \theta)$ be the small-fugacity Euler–Maclaurin remainder used in Section 5.1, with θ held fixed. For every fixed d ,

$$\partial_x^d R_P(x, \theta) = O_{a,b,P,d,C} \left(\frac{t^{2P-1}}{ux^d} \right), \quad |\theta| \leq Ct/L. \quad (13a)$$

This follows directly by differentiating its periodic-Bernoulli remainder integral: $v = 1/u$ has $v^{(r)} = O(vx^{-r})$, $t^{(r)} = O(tx^{-r})$, and the occurring weighted derivative integrals satisfy

$$\int_0^\infty y^k \left| \partial_w^\ell D_v^h \log(1 + ve^{-w}) \right|_{w=yz} dy = O(v/t^{k+1})$$

for the nontrivial derivatives that occur. Factors from differentiating z^{2P} and from $yt'(x)$ supply precisely the displayed powers. Moving arc endpoints contribute only $e^{-c/t}$ -scale boundary terms. On minor arcs, product differentiation removes at most d factors and costs only fixed polynomial factors. Consequently errors can still be made smaller than any prescribed algebraic order after any fixed number of derivatives; the unchanged exact Δ_{K+1} rate is not asserted for these differentiated errors. In particular,

$$G_\alpha(x) = \log \mathcal{A}_\alpha(x), \quad G'_\alpha(x) \sim \frac{\alpha(\log x + 2)}{\sqrt{2x}} > 0. \quad (14)$$

For an explicit check of the first derivative, write $z = t - i\theta$, $F = F(e^{-z}, u)$, and differentiate at fixed t before accounting for $t = t(x)$. Integration by parts in θ gives the exact identity

$$\mathcal{A}'_\alpha(x) = \frac{1}{2\pi} \int_{-\pi}^\pi e^{xz} \left\{ zF + \frac{\alpha u}{x} F_u \right\} d\theta + \frac{t'(x)}{\pi} e^{xt} F(-e^{-t}, u) \sin(\pi x). \quad (14a)$$

The last term is a minor-arc boundary term and is beyond every power relative to the main term. On the principal arc, the leading multiplier is $z + \alpha \log(1 + u)/(xz)$, since $uA'(u) = \log(1 + u)$. Its saddle value is $t + \alpha \log(1 + u)/(xt)$, giving (14). This argument uses the derivative of the product itself on the minor arcs, so possible zeros of F cause no division problem.

Thus (13) is eventually strictly increasing and has a well-defined large positive inverse.

If X is the target and $y = \log X$, set

$$z = \frac{y}{2\sqrt{2\alpha}}, \quad r_0 = \frac{z}{W_0(z)}, \quad x_0 = r_0^2. \quad (15)$$

Then $\mathcal{A}_\alpha^{-1}(X)$ is asymptotic to x_0 . Writing $l_0 = \log x_0$, the first explicit inverse correction is

$$\mathcal{A}_\alpha^{-1}(X) = x_0 - \frac{\pi^2 x_0}{3\alpha^2 l_0(l_0 + 2)} + O_{a,b}(x_0/l_0^4). \quad (15a)$$

Indeed $G(x) - g(x) = \frac{\sqrt{2\pi^2}}{6\alpha} \frac{\sqrt{x}}{\log x} + O(\sqrt{x}/\log^3 x) + O(\log x) + O(x^{1/2-\alpha}/\log x)$. Divide the first term by $g'(x)$, and apply Taylor's theorem. The quadratic displacement and the next phase term are

$O(x/\log^4 x)$; the prefactor and finite- u corrections are smaller uniformly for $\alpha \geq a > 0$. This is a first logarithmic inverse transseries term, not an absolute-error certificate.

For controlled finer corrections let $S_{K,M}(x)$ be the finite sum in (3), and solve

$$\log S_{K,M}(x) = y$$

by Newton iteration beginning near x_0 . The approximant is positive and increasing eventually; its derivative has the same leading equivalent (14). If x_* is its large solution and x is the exact inverse of (13), then

$$x - x_* = O\left(\frac{\sqrt{x_*}}{\log x_*} \left\{ \frac{t(x_*)^{M+1}}{x_*^\alpha} + e^{-\Delta_{K+1}(x_*)} \right\}\right). \quad (16)$$

This is an arbitrary-fixed-algebraic-order inverse procedure when M is increased. For α below $1/2$, the exact dilogarithm must remain in $\mathbb{S}$; replacing it by A_∞ before inversion discards an unbounded logarithmic correction.

A formal inverse coefficient generator is also available. Let $g(x) = \sqrt{2}\alpha\sqrt{x}\log x$, let $H = g^{-1}$ be given by (15), and put $E(x) = \log S_{K,M}(x) - g(x)$, $\eta(y) = E(H(y))$. Lagrange reversion in an auxiliary parameter multiplying E gives

$$x = H(y) + \sum_{r \geq 1} \frac{(-1)^r}{r!} \left(\frac{d}{dy}\right)^{r-1} \{H'(y)\eta(y)^r\}. \quad (17)$$

Equation (17) is a formal reversion generator; truncating it requires the usual ordered error bookkeeping. A numerical Newton solve with (16) avoids asserting convergence of an infinite transseries. Every derivative includes $u = x^\alpha$, $A(u)$, $N(x)$, and the finite-sector dependence.

7.1 Integer threshold

The distinct-partition map that increments the largest part embeds partitions of n into those of $n+1$ without changing part count. Since $(n+1)^\alpha > n^\alpha$, the weighted sequence is strictly increasing for $n \geq 1$. For sufficiently large X , because (13) is monotone and agrees with the sequence at integers,

$$\min\{n : a_n(\alpha) \geq X\} = \lceil \mathcal{A}_\alpha^{-1}(X) \rceil.$$

An approximate inverse must **not** simply be rounded when its error interval crosses an integer. Use an enclosure from a certified remainder, or evaluate the neighboring exact coefficients. The big-O constants in this note are not numerical certificates for a particular finite input.

[write] **An instance of the repository's inversion apparatus.** No novelty is claimed here for the inversion method; the instances are those of the transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`). The initializer (15) solves $r \log r = z$, that is $X + \log X = \log z$ for $X = \log r = \frac{1}{2} \log x$: the dominant block of `p0:thm:lambert-core` with $a = b = 1$, on the principal branch, so $r_0 = e^{W_0(z)} = z/W_0(z)$. The generator (17) is the reversion formula of `p0:thm:perturbed-inversion`, read formally at $\varepsilon = 1$ with $F = g$, $\mu_0 = H$ and $E(\mu_0(y)) = \eta(y)$; the manuscript itself calls it formal and does not assert convergence. The threshold

identity of this subsection is part (1) (exact rounding) of `p0:thm:staircase` for the interpolation (13), and the warning against rounding an approximate inverse is its separation condition, part (2).

8 Exact computations and practical warning

The code `checks.py` computes the exact polynomials

$$[q^n]F(q, u) = \sum_{m \geq 1} p_{\leq m}(n - m(m+1)/2)u^m$$

using the standard restricted-partition dynamic program. For $\alpha = 1, 2$ the evaluation is exact integer arithmetic, followed by 100-digit comparisons. For fractional α the polynomial coefficients remain exact and evaluation uses 100-digit arithmetic. The displayed low $\alpha = 2$ values agree term for term with OEIS through $n = 23$; $\alpha = 1$ gives 1,1,2,12,20,55,294,497,1224,... .

At $n=20,000$:

- $\alpha = 1$: principal Bessel relative error is about -9.0139×10^{-5}
- $\alpha = 2$: principal Bessel relative error is about -5.5771×10^{-2}
- $\alpha = 1/4$: principal relative error is about -2.9187×10^{-10} , reduced to about 2.24×10^{-16} by the c_2 correction

Adding $c_2, c_3, \dots$ does essentially nothing to the first two errors because the oscillatory sectors dominate the algebraic endpoint correction at these sizes. This late onset is explained, rather than contradicted, by (4).

`resonance_check.py` compares finite conjugate Bessel sectors, keeping endpoint degree $M = 9$. At $n=20,000$:

- $\alpha = 1$: $K = 0$ gives -9.0139×10^{-5} , $K = 1$ gives $+4.43 \times 10^{-9}$, $K = 2$ gives -4.92×10^{-12}
- $\alpha = 2$: $K = 0$ gives -5.5771×10^{-2} , $K = 1$ gives -1.0985×10^{-2} , $K = 2$ gives -5.09×10^{-5}

These numerical improvements are diagnostics, not certified finite- n error bounds. Increasing K indefinitely is unsafe: (3) fixes K before n tends to infinity, and the formal infinite Bessel sum need not converge. Remote root-of-unity arcs and modular factors in (11) can matter before the eventual regime. The script records all K from 0 through 8 so this limitation is visible rather than hidden.

9 Prior-art boundary and sources checked

- OEIS A291698: <https://oeis.org/A291698>. Sequence definition and leading conjecture.
- OEIS A292304: <https://oeis.org/A292304>. Sequence definition and leading conjecture.
- Vivien Brunel, *A general asymptotic formula for distinct partitions*, arXiv:1709.04955, journal DOI 10.1016/j.aop.2018.03.023: <https://arxiv.org/pdf/1709.04955>
- Dan Romik, *Partitions of n into $t\sqrt{n}$ parts*, European Journal of Combinatorics 26 (2005), 1–17, DOI 10.1016/j.ejc.2004.02.005: <https://kam.mff.cuni.cz/~ksemweb/clanky/Romik-partitions.pdf>
- Arash Arabi Ardehali and Hjalmar Rosengren, *A New Product Formula for $(z; q)_\infty$, with Applications to Asymptotics*, published 19 September 2026: <https://link.springer.com/article/10.1007/s00365-026-09783-2>

- Richard J. McIntosh, *Some asymptotic formulae for q -shifted factorials*, Ramanujan Journal 3 (1999), 205–214: <https://doi.org/10.1023/A:1006949508631> (bibliography verified in the 2026 primary paper; this 1999 full text was not reviewed here)
- NIST DLMF: Jacobi triple product (Chapter 17), θ transformations (Chapter 20), and modified Bessel large-argument asymptotics (10.40): <https://dlmf.nist.gov/17.8> ; <https://dlmf.nist.gov/20.7> ; <https://dlmf.nist.gov/10.40>

Brunel’s Section 3 treats prescribed part count by a two-variable saddle and recovers classical regimes; it does not state the polynomial moving-fugacity all-order remainder or finite conjugate-sector theorem above. Romik’s Theorem 2 is uniform for the normalized part count on compact subsets strictly below the distinct-partition endpoint $\sqrt{2}$; polynomial fugacity approaches that endpoint. His local-limit proof is relevant methodological prior art. These checks delimit those two sources; they do not prove a global novelty claim, and the grand-canonical/Fermi-gas interpretation and leading saddle method are classical.

The September 2026 paper supplies a uniform moving-argument q -product expansion (Corollary 2) and discusses the classical large-argument theta regime. It does not state the coefficient-diagonal or inverse theorem studied here. Its Corollary 2 assumes a fixed argument for the small complex parameter and a lower bound on the real part of the transformed argument. These are relevant antecedents, not a basis for claiming that q -product Euler–Maclaurin or modular transformations are new.

[write] **The two OEIS conjectures.** The delivered proof note (`theorem-and-proof.md`, Section 9) records that the OEIS page of A291698 could not be reached when the manuscript was written, so that its displayed conjecture was transcribed from a screening note that is not part of the delivery, while the A292304 entry was checked directly. The manuscript’s list above drops that caveat. On 2 October 2026 both entries were consulted for this write: the equivalents displayed in Section 6 are exactly the conjectures of Václav Kotěšovec in A291698 (15 September 2017) and A292304 (14 September 2017), which the manuscript does not name; and the listed terms 1, 1, 2, 12, 20, 55, 294, 497, 1224, 2520 of A291698 agree with an independent recomputation.

[write] **Relation to other reports of the collection.** The neighbouring report `a022629-distinct-partition-norms`, in the same directory as this one, treats the norm-weighted products $\prod_{k \geq 1} (1 + k^\alpha q^k)^\mu$ (A022629 and its relatives), where the weight k^α depends on the *part* k . Here the weight n^α depends on the *size* n being counted and is the same for every part, so a partition with m parts carries $n^{\alpha m}$. The two families share the distinct-part structure, the Euler–Maclaurin and saddle methods and a Lambert-type inverse, but not the saddle regime (here the part count approaches the distinct-partition endpoint $\sqrt{2}n$, with conjugate resonance sectors governed by $L = \log u$), the sequences or any theorem. Neither manuscript cites the other. That report was not edited by this write.

10 Further research questions

10.1 Effective constants and certified numerical ranges

The present bounds are uniform but eventual. An explicit value for the consecutive-interval constant, quantitative Euler–Maclaurin remainders, and certified saddle-contour quadrature would turn the inverse enclosure into a finite-input certificate. This is especially relevant for $\alpha = 2$, where asymptotic smoothness begins late. The exact local integrals in (12c) provide a natural numerical starting point; the supplied quadrature experiment is not interval arithmetic.

10.2 A growing number of resonance sectors

The proof fixes K and an auxiliary J before sending n to infinity. Extending it to $K = K(n)$ requires uniform control of the changing saddle geometry and the modular factor $(Q; Q)_\infty^{-1}$. Near remote root-of-unity arcs that factor need not be close to one. An unqualified infinite sum of the formal Bessel sectors is not justified and may fail to converge.

10.3 The transition beyond polynomial fugacity

The first resonance scale is governed by $\sqrt{n}/L^3$, where $L = \log u$. Polynomial fugacity has $L \asymp \log n$, so every fixed nonprincipal sector is beyond all algebraic orders. A different regime should appear when L is comparable to $n^{1/6}$, for which that damping is of order one. The present theorem does not cover that regime. A uniform theta-type crossover would need its own proof rather than an assumption that a Gaussian approximation remains accurate for a small part-count variance.

[write] **Added 5 October 2026, batch 100.** A neighbouring model, not the regime asked for here, is treated in Part II of the report `oeis-sequence-asymptotics/a271619-strict-twice-partitions` (Report 181 of the session bundle; labels `stp:len:`): A358836, $\prod_{k \geq 1} (1 + q^k/(q; q)_k)$, a distinct-part product whose fugacity $1/(q; q)_k \rightarrow P(q)$ depends on q itself, with $\log u \asymp A/t \asymp n^{1/4}$ at its saddle. Its all-angle bound (Lemma 19.1 there) leaves no resonance sector, and its source audit says substitution into Theorems 1–2 here is not justified, so the regime $\log u \asymp n^{1/6}$ of this subsection remains open. Its expansion of $\log \prod_k (1 + e^{\mu(z)} q^k)$ (equation (96) there) is identity (11) of this report at the complex fugacity $u = e^{\mu(z)}$, derived there independently, with a dual-term bound on a moving complex disk of radius $\asymp t^2$.

10.4 Comparison with general q-product asymptotics

The endpoint expansion and modular ingredients are classical. The appropriate contribution to compare against the literature is the diagonal coefficient transfer with moving fugacity, explicit finite exponential sectors, and controlled inversion. The recent moving-argument product formulas cited in Section 9 may simplify effective remainder calculations. A broader search is still needed before making a publication-level novelty claim.

A Reproducibility and independent checks

The reproducibility package contains the complete editable L^AT_EX source, the proof-note source, four Python programs, and their saved numerical output. The mathematical source used for integration has SHA256

5a90f1336fbbd5c0804f4781f436d47cb18a47ef82cefc04ef1f322d5cc9f64b

The long hash identifies the source precisely; it is not a mathematical certificate.

A.1 Programs and outputs

checks.py

Generates exact restricted-partition polynomials, the endpoint coefficient recurrence, principal Bessel comparisons through $n = 20,000$, and principal-model inverse checks.

Output: checks.json.

resonance_check.py

Evaluates the finite conjugate Bessel sums for $K = 0, \dots, 8$, with $M = 9$.

Output: resonance-checks.json.

validate.py

Performs 120 independent direct weighted-product dynamic-programming checks for $n \leq 60$, verifies 24 listed A292304 terms, tests the exact Jacobi identity at real and complex arguments, and compares finite-sector inverses.

Output: validation.json.

sector_integrals.py

Performs uncertified high-precision quadrature of the exact local integrals at $\alpha = 1$, $n = 20,000$, for $j = 0, 1, 2, 3$.

Output: sector-integral-checks.json.

[write] **Shipped names.** In this repository the four programs are `code/checks.py`, `code/resonance_check.py`, `code/validate.py` and `code/sector_integrals.py`, their outputs are in `data/`, the delivered `build_pdf.sh` is `code/build_pdf.sh`, and `moving-fugacity-report.tex` is `article.tex`. The last three programs read and write files beside themselves, so the replay below must be run on a flat copy; the report's README gives the commands.

The checks were run with Python, mpmath 1.3.0, and SymPy 1.14.0. The principal comparison uses 100 decimal digits; the local-sector quadrature uses 70. All integer polynomial coefficients are exact. No random seed is involved.

A.2 Clean replay

From the package directory, run

```
python checks.py --max-n 20000 --output checks.json
python resonance_check.py
python validate.py
python sector_integrals.py
pdflatex -interaction=nonstopmode moving-fugacity-report.tex
pdflatex -interaction=nonstopmode moving-fugacity-report.tex
```

The JSON values are high-precision numerical observations, not rigorously rounded enclosures. The proof does not depend on their being small.

A.3 Finite-sector inverse diagnostics

For target $y = \log a_{20000}(\alpha)$, Newton inversion of the $M = 9$ model gives the following signed errors $x_* - 20000$:

α	$K = 0$	$K = 1$	$K = 2$
1	1.49946×10^{-3}	-7.36613×10^{-8}	8.18152×10^{-11}
2	4.81105×10^{-1}	8.31548×10^{-2}	3.80911×10^{-4}

These values demonstrate the importance of the first conjugate sectors. They do not certify a ceiling when the true inverse lies at an integer: the exact coefficient or an error enclosure must still decide the threshold.

A.4 Identity and local-integral diagnostics

Three direct evaluations of the exact Jacobi identity had relative discrepancies below 3×10^{-77} at the selected working precision. Independent direct weighted-product evaluation agreed exactly with the restricted-partition-polynomial method in all 120 small cases. All 24 listed A292304 terms agreed exactly.

At $\alpha = 1$, $n = 20,000$, the $M = 9$ expansion matched the numerically integrated principal local sector to about 1.28×10^{-28} relative, and the first complex sector to about 3.61×10^{-27} . Higher fixed cells at this finite size had larger endpoint effects; no monotonic improvement in sector count is promised. These checks support the coefficient signs and normalizations but are not substitutes for the contour proof or certified quadrature.

B Scope checklist

- Uniformity is on a fixed positive compact range of α .
- The sector count K , auxiliary count J , and endpoint depth M are fixed before n tends to infinity.
- The exact amplitude is deformed only within $\Re z > 0$; only truncated polynomial integrands are completed around a full circle.
- Exact dilogarithms are retained for general positive α ; the elementary Sommerfeld equivalent is relatively valid only for fixed $\alpha \geq 1/2$.
- The real inverse refers to the explicitly defined saddle–Fourier continuation.
- Big- O constants in the report are not finite-input numerical certificates.
- No external submission, OEIS edit, or publication-priority assertion is part of this report.